

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14 (Ancillary)

Paper No. BSC-06A: Ancillary Mathematics (Old Course)

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Draw the graph of the function $y = e^x$.
- (b) Find $\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$.
- (c) Find $\frac{d}{dx}(\log \sqrt{x})$.
- (d) Find $\int (ax^2 + 4x^3) dx$.
- (e) Define the order and the degree of a differential equation.
- (f) If $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 0 & 5 \end{pmatrix}$, then find the transpose of A .
- (g) If $A = \begin{pmatrix} 1 & 1 \\ 3 & 0 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 2 \\ -2 & 0 \end{pmatrix}$, find $2A - B$.
- (h) Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$. Find AB .
- (i) Define lower and upper triangular matrices.
- (j) Solve: $\frac{dy}{dx} = \frac{x+y}{x}$.
- (k) Define diagonal and scalar matrices.

Question 2. Find the following:

[12]

- (a) $\frac{d}{dx}(2^x)$
- (b) $\frac{d}{dx}(x^n \cos x)$
- (c) $\frac{d}{dx}(\sin \sqrt{x})$
- (d) $\frac{d}{dx}(x \log x)$

Question 3. Find the following: [12]

(a) $\int (1 - 3x)(1 + x) dx$

(b) $\int \frac{3}{x} dx$

(c) $\int_0^1 \frac{1}{x^2+1} dx$

(d) $\int xe^x dx$

Question 4.

(a) Find the turning points on the curve $y = x^3 - 3x^2 + 3x$ and distinguish them. Also find the maximum and minimum values of the function. [6]

(b) Show that $y = \frac{c}{x}$ is a solution of the differential equation $x \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$. [6]

Question 5.

(a) Find the area between the curve $y = x(4 - x)$ and the x -axis from $x = 0$ to $x = 5$. [6]

(b) Show that $\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1 + a + b + c$. [6]

Question 6.

(a) Compute the inverse of the following matrix:

$$\begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 0 \\ 1 & 4 & 1 \end{pmatrix}$$

[6]

(b) If $A = \begin{pmatrix} 3 & -4 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, show that $(AB)' = B'A'$. [6]

Question 7.

(a) Solve the following system of equations by matrix method:

$$2x_1 - x_2 + 3x_3 = 9$$

$$x_1 + x_2 + x_3 = 6$$

$$x_1 - x_2 + x_3 = 2$$

[6]

(b) Solve the following system of equations by Cramer's rule:

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

[6]